PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 5.3 (Federated Continual Learning under Drift)

Report prepared 19 August 2026

1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 5.3 (Federated Continual Learning under Drift) of the review, comprising all 14 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to Section 5.3's specific review question, namely federated continual learning under drift in IoT intrusion detection, with particular attention to the manuscript's own analytical lens: whether an article distinguishes genuine client-local concept drift from standing non-IID heterogeneity and from adversarial or Byzantine manipulation, a three-way discrimination problem the manuscript identifies as unaddressed across the literature it reviews. An article can present a clear problem statement for a related but distinct question, such as general federated-learning-IDS cataloguing, non-IoT domain-incremental vision, or heterogeneity-only or attack-only robustness, and still score partially or zero on QC1 here, because that problem statement is not the specific one this subsection screens for.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 5.3 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 14 full-text articles assessed for Section 5.3 against the quality criteria in Section 2, 8 articles met or exceeded their applicable threshold and were retained, while 6 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 5.3 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	Advances and open problems in federated learning (2021)	2021	1.0	1.0	1.0	0.5	**3.5**	≥2.5	**Include**
2	Federated continual learning: A comprehensive survey on lifelong and privacy-preserving learning over distributed and non-stationary data (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
3	Non-IID data and continual learning processes in federated learning: A long road ahead (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
4	FD-IDS: Federated learning with knowledge distillation for intrusion detection in Non-IID IoT environments (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
5	FL-MalDrift: A federated learning framework for malware detection under local concept drift (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
6	ATTA-FL-Lite: Lightweight byzantine-robust federated learning for resource-constrained medical IoT devices (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
7	Secure and explainable federated learning for IoT intrusion detection: A comprehensive survey (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
8	XDFC-IDS: An explainable decentralized federated class-incremental fusion framework for intrusion detection (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
9	Federated learning in intrusion detection: advancements, applications, and future directions (2025)	2025	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
10	Fed-CALiBER: Federated Lightweight BERT Intrusion Detection on CAN Bus Protocol in Autonomous Vehicle (2025)	2025	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
11	Federated continual learning with domain-routed historical expert matching for multimodal perception (2026)	2026	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
12	Adaptive Hierarchical Federated Learning for IoT Anomaly Detection (2025)	2025	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
13	Non-IID and aware federated intrusion detection with PBFT with secured model aggregation for multi institutional healthcare internet of things networks (2026)	2026	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
14	Communication-Efficient Federated Class-Incremental Intrusion Detection for Edge IoT Networks (2026)	2026	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**

This retained set of 8 articles constitutes the eligible pool for Section 5.3 (Federated Continual Learning under Drift) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

Prepared by
Name: RAZIMAN ZAKARIA